

methods and rates are still supported for backwards compatibility.

Operating frequency band. Table II shows operating frequency bands for different Wi-Fi protocol standards. Some Wi-Fi modules support dual frequency bands. For example, while 802.11n solution supports both the 2.4GHz and 5GHz bands, 802.11g supports only 2.4GHz band. Each range is divided into a multiple channels. Countries apply regulations on number of channels as well as users and maximum power levels within a frequency band.

Data rate. Each Wi-Fi standard is rated according to its maximum theoretical data rate. Different standards have different data rates due to the technology used for each standard. Practically, these data rates are never reached. These are only the theoretical maximum value for a particular technology. The theoretical maximum data rate of a Wi-Fi module could be from 1Mbps (802.11b) to 6.75Gbps (802.11ac). You need to decide the required data rate for your application. Higher data rate would not always be the best choice as sometimes high data rates would not even be sustainable with large number of clients.

Transmit range. The range of a Wi-Fi network is always limited and majorly depends on the specific protocol used and the transmission power. Interference and obstruction from environment also plays some role. For example, approximate maximum indoor range for 802.11g is around 38 metres.

OS support. It is highly desirable that the module quickly gets connected and configured with Android, iPhone, Linux and other embedded OS. If you need to build in this feature into your product, you need to check the driver level support, and configuration and management support from the Wi-Fi module carefully.

Antenna. Both omni-directional and directional antennae are available in Wi-Fi modules. Omni-directional antennae are typically used for wireless coverage providing connectivity

TABLE I
Different Manufacturers of Wi-Fi Modules

Redpine Signals	Digi International
Rabbit Semiconductor	Wi2Wi
Lantronix	Rabbit Semiconductor
Laird Technologies	Silicon Laboratories
Microchip	Infineon
Murata	Freescale
B&B Electronics	Atmel

TABLE II
Standards and Corresponding Frequency Bands

Frequency Band	Standard
2.4GHz (2.4 - 2.483 GHz)	802.11b/g/n
5GHz (5.15 - 5.725 GHz)	802.11a/h/j/n/ac
5.9GHz (5.85 - 5.9 GHz)	802.11p

to Wi-Fi devices inside office, home, warehouse, etc. Directional antennae on the other hand are used for focusing the wireless signal in a specific direction, resulting in a limited coverage area. The typical applications are wireless connection from one building to another. Through use of directional antennae, the connection can be extended with many kilometres between stations.

Secure Wi-Fi authentication schemes. In 2001, a group from the University of California, Berkeley presented a paper describing weaknesses in the 802.11 Wired Equivalent Privacy (WEP) security mechanism defined in the original standard. The IEEE then set up a dedicated task group to create a replacement security solution. The Wi-Fi Alliance announced an interim specification called Wi-Fi Protected Access (WPA) based on a subset of the then current IEEE 802.11i draft. WEP, WPA, WPA2, WPA2-Enterprise, WPS, WMM, WMM-PS are typical Wi-Fi security types these days. Each type has different advantages and disadvantages. These security schemes help protect information from Wi-Fi devices on a network.

Packaging type. Both surface-mount (SMT) and through-hole pack-

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